UNIVERSITY OF BARISAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

3rd Year 2nd Semester Final Examination

Course Title: Peripherals and Interfacing

Course Code: CSE-3207

Time: 3 hours **Marks:** 60

Answer any five Questions from the followings.

1. a) What is Interface? Briefly explain different types of Interface [5]
   b) What do you mean by Computer peripheral? List down the input and output Peripherals. [2]
   c) Differentiate between off-line and on-line peripherals [2]
   d) Write short note on Universal Serial Bus [3]
2. a) Define the terms **data path** and **control path**. Explain their roles in the execution of instructions and how they interact with each other within a CPU [4]
   b) Explain the components and functionality of an **adder** and a **multiplexer** in digital circuits. [4]
   c) Define **control hazards**, **data hazards**, and **structural hazards** in the context of pipelined CPU architectures. Discuss the impact of each type of hazard on instruction execution and the techniques used to mitigate them. [4]
3. a) What do you mean by the accuracy of an optical mouse? Describe the factors that affect the quality of an optical mouse. [4]
   b) Write short notes on: [4]
      i. Polling rate
      ii. Refresh rate
   c) Can a mouse's sensitivity (DPI) be adjusted, and why might someone want to change it? [4]
4. a) Describe the function and importance of the Arithmetic Logic Unit (ALU) in a CPU. What types of operations does the ALU perform, and how does it interact with other components of the processor during instruction execution? [4]
   b) Create a flowchart to illustrate the process of multiplying two numbers using assembly language. Explain the key steps involved in the flowchart and how they translate into assembly language instructions. [4]
   c) Define the terms **Stack Pointer (SP)** and **Stack Memory** in the context of computer architecture. [4]
5. a) Define **cycle time** and **CPI (Cycles Per Instruction)**. How do these metrics influence processor performance? Provide an example of how reducing CPI or cycle time can improve overall performance. [6]
   b) Define registers in the context of computer architecture. What are the different types of registers, and how do they contribute to the overall performance of a CPU? [6]
6. a) Define processor architecture and explain its importance in computer systems. Describe the key components of a processor architecture and how they work together to execute instructions. [6]
   b) Compare RISC (Reduced Instruction Set Computing) and CISC (Complex Instruction Set Computing) architectures. What are the advantages and disadvantages of each, and in what types of applications are they typically used? [6]
7. a) Explain the architecture and performance of GPU. Why are GPUs particularly well-suited for parallel processing tasks? [4]
   c) Discuss the advancements introduced with Bluetooth 4.0 technology in comparison to its predecessors. What key features made Bluetooth 4.0 a significant improvement for mobile devices. [4]
   d) USB 3.0, launched in 2010, introduced key improvements over earlier versions. Explain its main features and benefits. [4]
8. a) Write short notes on the Hall effect keyboard. [4]
   b) Write down the difference between laser keyboard and dome switch keyboard [4]
   c) Discuss the impact of proper keyboard ergonomics on user productivity and health. What are some key ergonomic tips for reducing strain when using a keyboard for long periodspp? [4]

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